



Red and Processed Meat, Poultry, Fish, and Egg Intakes and Cause-Specific and All-Cause Mortality among Men with Nonmetastatic Prostate Cancer in a U.S. Cohort

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ABSTRACT

Background: Research on the relationship of meat, fish, and egg consumption and mortality among prostate cancer survivors is limited.

Methods: In the Cancer Prevention Study-II Nutrition Cohort, men diagnosed with nonmetastatic prostate cancer between baseline in 1992/1993 and 2015 were followed for mortality until 2016. Analyses of pre- and postdiagnosis intakes of red and processed meat, poultry, fish, and eggs included 9,286 and 4,882 survivors, respectively. Multivariable-adjusted RRs and 95% confidence intervals (CI) were estimated using Cox proportional hazards models.

Results: A total of 4,682 and 2,768 deaths occurred during follow-up in pre- and postdiagnosis analyses, respectively. Both pre- and postdiagnosis intakes of total red and processed meat were positively associated with all-cause mortality (quartile 4 vs. 1: RR = 1.13; 95% CI, 1.03–1.25; $P_{\text{trend}} = 0.02$; RR = 1.22; 95% CI, 1.07–1.39;

$P_{\text{trend}} = 0.03$, respectively), and both pre- and postdiagnosis poultry intakes were inversely associated with all-cause mortality (quartile 4 vs. 1: RR = 0.90; 95% CI, 0.82–0.98; $P_{\text{trend}} = 0.04$; RR = 0.84; 95% CI, 0.75–0.95; $P_{\text{trend}} = 0.01$, respectively). No associations were seen for prostate cancer-specific mortality, except that higher postdiagnosis unprocessed red meat intake was associated with lower risk.

Conclusions: Higher red and processed meat, and lower poultry, intakes either before or after prostate cancer diagnosis were associated with higher risk of all-cause mortality.

Impact: Our findings provide additional evidence that prostate cancer survivors should follow the nutrition guidelines limiting red and processed meat consumption to improve overall survival. Additional research on the relationship of specific meat types and mortality is needed.

Introduction

Over 3.3 million prostate cancer survivors currently reside in the United States (1) and an additional 174,650 men are expected to be diagnosed in 2019 (2). Diet after prostate cancer diagnosis is an important concern to prostate cancer survivors. A previous pooled analysis of 15 cohort studies including the Cancer Prevention Study-II (CPS-II) Nutrition Cohort suggests that red and processed meat consumption is associated with moderately higher risk of advanced prostate cancer; poultry is inversely, and eggs are positively associated with both advanced and fatal prostate cancer risk; and seafood is not associated with prostate cancer regardless of stage or grade (3). Cancer survivors are advised to follow nutrition guidelines for cancer prevention, which include limiting red and processed meat intake (4). However, little is known about the associations between meat and egg intakes after cancer diagnosis and cause-specific and overall mortality among prostate cancer survivors.

Four prospective analyses among three cohort studies examined meat consumption at or after cancer diagnosis in relation to prostate cancer progression (5–8), with 3 defining progression as primarily

biochemical recurrence (2 or more consecutive PSA concentrations ≥ 0.2 ng/mL) along with other events and only one as metastases or prostate cancer-specific mortality (PCSM), which is more relevant to survival (8). Results were null for total red and processed meat intake and recurrence/progression (5–7), but the one study that examined metastasis or PCSM found a suggestive positive (not statistically significant) association for postdiagnosis processed meat consumption (8). Results were also inconsistent for poultry, fish, and egg intakes (5–8). Small numbers of cases, inconsistent definitions of prostate cancer progression, and short follow-up times may partially explain the inconclusive findings.

Associations of prediagnosis diet with mortality outcomes after prostate cancer diagnosis are also potentially informative because diet before diagnosis is less subject to reverse causation bias than postdiagnosis diet, and prediagnosis diet might play a role in the development of tumor aggressiveness. One study examined prediagnosis fish intake among prostate cancer survivors and found a significant inverse association with PCSM (9). No previous studies have examined prediagnosis intakes of red and processed meat, poultry, or eggs with cause-specific and all-cause mortality among prostate cancer survivors.

We examined the associations of both pre- and postdiagnosis intakes of total red and processed meat, unprocessed red meat, processed meat, poultry, fish, and eggs with PCSM, cardiovascular disease-related mortality (CVD), and all-cause mortality among men diagnosed with nonmetastatic prostate cancer in the CPS-II Nutrition Cohort.

Materials and Methods

Study population

Participants in this study were drawn from 86,402 men enrolled in the CPS-II Nutrition Cohort, a prospective study of cancer incidence and mortality, established by the American Cancer Society in 1992 as

Behavioral and Epidemiology Research Group, American Cancer Society, Atlanta, Georgia.

Note: Supplementary data for this article are available at Cancer Epidemiology, Biomarkers & Prevention Online (<http://cebp.aacrjournals.org/>).

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described in detail elsewhere (10). At enrollment, participants completed a mailed, self-administered 10-page questionnaire. Follow-up questionnaires were sent to living participants in 1997 and every other year to update medical and lifestyle information and ascertain newly diagnosed cancers. The Nutrition Cohort has been approved by the Institutional Review Board at Emory University (Atlanta, GA).

We identified 11,734 prostate cancer cases diagnosed between enrollment in 1992/1993 and 2015. We excluded 227 prostate cancer deaths that could not be verified through medical records or linkage with state cancer registries. We also excluded men with implausible diagnosis dates ($n = 40$), unknown stage ($n = 367$), distant stage ($n = 340$), non-adenocarcinoma histology ($n = 6$), or prostate tumors classified as carcinoma *in situ* ($n = 6$), resulting in a total 10,748 men potentially eligible for pre- and postdiagnosis dietary analysis (Fig. 1).

In prediagnosis analysis, we further excluded men whose dietary intake was not reliable [energy intake $>4,000$ kcal/day or <650 kcal/day or had more than 10 missing line items on the 68-item Block food frequency questionnaire (FFQ), $n = 927$, 8.6%]. We also excluded men whose FFQ was returned within one year before diagnosis to avoid influence of prediagnostic symptoms on diet ($n = 535$, 5%). In postdiagnosis analyses, we excluded from the 10,748 eligible men those whose postdiagnosis dietary intake was not available because they were diagnosed with prostate cancer after 2003 (diet was not assessed after 2003; $n = 3,416$, 31.8%) or, for those diagnosed before 2003, dietary intake was missing/unreliable (unreliable FFQ was defined as energy intake $>4,200$ kcal/day or <800 kcal/day or had more than 70 missing line items on the 152-item Willett FFQ; $n = 1,601$, 14.9%). We further excluded men who returned the postdiagnosis survey within one year of diagnosis to avoid influence of treatment on diet ($n = 621$, 5.8%). Furthermore, deaths that occurred within 2 years of completion of postdiagnosis survey were excluded to minimize bias due to reverse causation ($n = 228$, 2.1%). After all exclusions, a total of 9,286 and 4,882 prostate cancer cases were included in the pre- and postdiagnosis dietary analyses, respectively.

Diet assessment

Prediagnosis dietary intakes of unprocessed red meat, processed meat, fish, poultry, and eggs were assessed at baseline in 1992/1993 from a validated, modified brief Block FFQ including 68 items (11, 12). Postdiagnosis dietary information was assessed from a modified 152-item Willett FFQ either in 1999 (for men diagnosed before 1999) or 2003 (for men diagnosed between 1999 and 2003, and men diagnosed before 1999 but missing 1999 FFQ), which was previously validated in similar populations (13, 14). Questions on red and processed meat, fish, poultry, and egg intakes were generally comparable across FFQs as shown in Supplementary Table S1. Dietary exposures were defined as follows: unprocessed red meat included beef, pork, lamb, and veal and excluded organs; processed meat included sausages, hot dogs, bacon, ham, and luncheon meats. Total red and processed meat was the sum of unprocessed red meat and processed meat. Poultry included chicken or turkey unprocessed meat and sandwiches, which may include processed white meat. Sandwiches were included to keep postdiagnosis poultry variable consistent with baseline poultry variable (sandwiches were not separable in 1992 FFQ). Fish included fish and shellfish, and eggs included only eggs with yolk.

Outcomes

All deaths were ascertained through linkage with the National Death Index through December 31, 2016 (15). The primary outcomes in this analysis were PCSM, defined as deaths with prostate cancer

coded as the singular underlying cause of death (International Classification of Disease Codes 9th revision: 185, 10th revision: C61) and all-cause mortality. We also examined CVD as the secondary outcome.

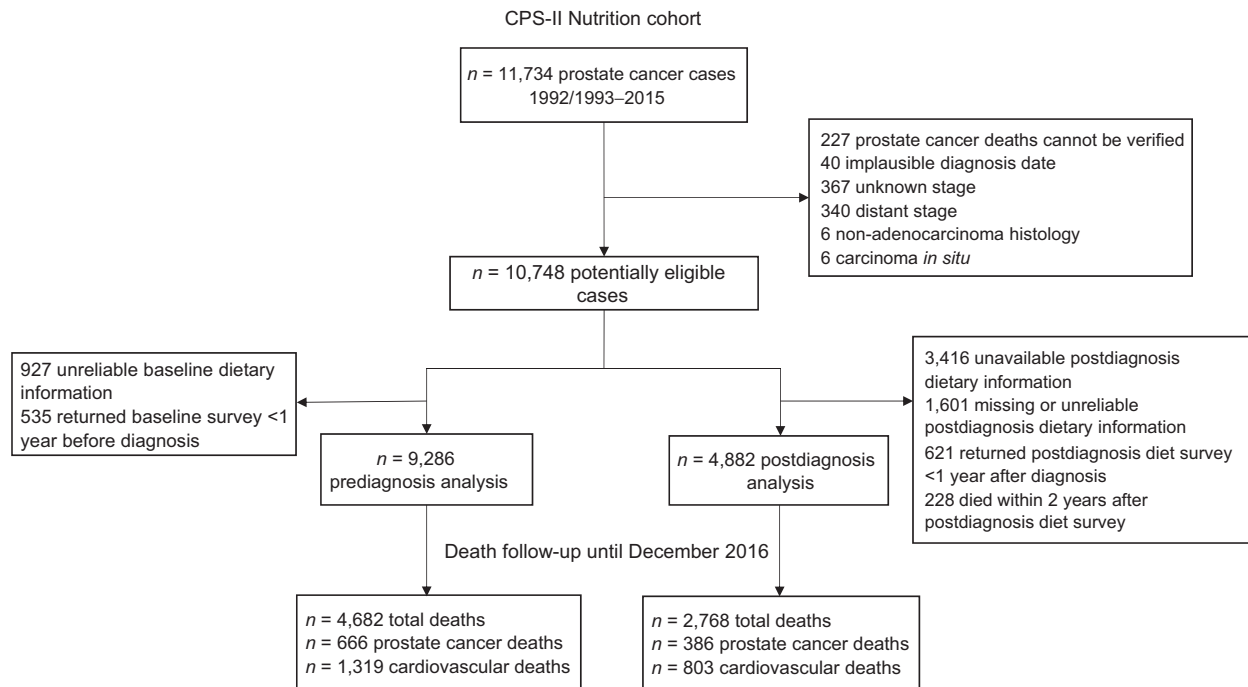
Statistical analysis

Questionnaire-specific quartiles were created for all exposures. Cox proportional hazards regression models were used to calculate RRs and 95% confidence intervals (CI) of cause-specific and overall mortality across quartiles of exposures. P values for trend were calculated when diet was modeled using the median of each quartile as a continuous variable. Follow-up time in prediagnosis analyses started from diagnosis date and ended on death date or December 31, 2016, whichever came first. Follow-up time in postdiagnosis analyses started 2 years after completion of the postdiagnosis questionnaire using a delayed-entry model and ended on death date or December 31, 2016, whichever came first. This 2-year lag was incorporated to minimize the influence of reverse causation and is consistent with our previous analysis on dietary exposures and prostate cancer survival (16).

All multivariable-adjusted models were stratified on single year of age at diagnosis and controlled for calendar year of diagnosis (1992–1998, 1999–2005, 2006–2014), tumor extent (T1–T2, T3–T4), nodal involvement (yes, no), and Gleason score (2–6, 7, 8–10, 5–7, 7–10, or unknown). The reason for these overlapping categories is that the only available Gleason score information for some cases was a categorical grade recorded in registry data, and these registry definitions of low, intermediate, and high grade varied over time (17). In prediagnosis multivariable models, we further adjusted for variables obtained at or derived for baseline, including education ($<$ high school, high school graduate, some college, $\geq$ college graduate), family history (limited to first-degree relatives) of prostate cancer (yes, no), any history of PSA testing not leading to prostate cancer diagnosis (yes, no, unknown), body mass index (BMI, kg/m^2 ; <18.5 , 18.5 – <25 , 25 – <30 , ≥ 30 , unknown), smoking status (never, current, former, unknown), recreational physical activity [metabolic equivalent task-hour/wk (MET-h/wk), <3.5 , 3.5 – <8.75 , 8.75 – <17.5 , ≥ 17.5], history of diabetes (yes, no), history of CVD (yes, no), history of other cancer (yes, no), total fruit and vegetable intake (in quartiles), and energy intake (kcal/day, continuous). In the model of individual exposure variables, other exposures were also adjusted. Red and processed meat consumption were mutually adjusted for each other.

In postdiagnosis multivariable models, we adjusted for the same covariates included in prediagnosis models except that covariates in postdiagnosis models were assessed from the postdiagnosis survey (except PSA screening, which was assessed from prediagnosis survey). We further controlled for the self-reported first course of treatment reported on a separate questionnaire after cancer diagnosis (prostatectomy, radiation, cryosurgery, hormone only/watchful waiting, unknown) in all postdiagnosis models.

We also conducted the following sensitivity analyses: (i) excluding processed poultry from the poultry variable in postdiagnosis analysis, (ii) including men whose prediagnosis diet was assessed from the baseline survey that was returned within a year of diagnosis, and (iii) excluding men with a history of CVD from the postdiagnosis dietary analyses. The following factors were evaluated as potential effect modifiers for all combinations of exposures and primary outcomes: age (≤ 70 vs. > 70 years), follow-up time (≤ 10 vs. > 10 years), tumor prognostic risk category based on National Comprehensive Cancer Network guidelines [lower-risk (T1–T2 and Gleason score ≤ 7) vs. high-risk (T3–T4, or Gleason score ≥ 8 , or nodal involvement); ref. 18], BMI

**Figure 1.**

Prostate cancer cases drawn from the CPS-II Nutrition Cohort 1992/1993–2015, and number of deaths identified up to 2016.

(<25 vs. ≥ 25 kg/m²), and smoking status (never vs. ever). *P* values for interaction were calculated using the likelihood ratio test comparing models with and without interaction terms between continuous exposures and categorical covariates (19).

We also considered a substitution analysis by substituting 1 serving/wk of one food item for 1 serving/wk of another, assuming that the total servings per week of red/processed meat, fish, poultry, and eggs are constrained to a certain level for each individual. We applied the “leave-one-out” model explained in detail elsewhere (20). Briefly, we created a total meat and egg variable (continuous in serving/wk) by summing the above mentioned 5 dietary variables together. The effect of substituting poultry for red/processed meat, as an example, while holding the total consumption of meat and eggs constant can be estimated from poultry variable by leaving red/processed meat out of the multivariable-adjusted model that includes poultry, fish, eggs, and the total meat and egg variable.

Results

Through 2016, 4,682 deaths occurred during follow-up among 9,286 men included in the prediagnosis analysis (666 due to prostate cancer, 1,319 due to CVD) and 2,768 deaths occurred among 4,882 men included in the postdiagnosis analysis (386 due to prostate cancer, 803 due to CVD). Men who consumed more total red and processed meat at baseline were more likely to have lower education level, have no PSA testing history, be overweight or obese, be current smokers, have more than 2 drinks per day, have a history of diabetes, have higher intake of energy and lower intake of fruits and vegetables, and were less likely to meet physical activity guidelines (≥ 8.75 MET-h/wk; **Table 1**). Similar patterns were seen for men with high intakes of eggs. Men included in the postdiagnosis dietary analyses were slightly younger than men in prediagnosis analyses (mean age at diagnosis $\pm$ SD: 69.2 $\pm$

5.4 years vs. 72.1 $\pm$ 6.1). Similar distributions of the characteristics were seen among men who reported higher red and processed meat consumption on their postdiagnosis survey, except that they were more likely to have a family history of prostate cancer and consumed slightly more fruits and vegetables (Supplementary Table S2).

In the prediagnosis analysis, men in the top quartile of total red and processed meat intake had higher risk of all-cause mortality compared with men in the bottom quartile (RR = 1.13; 95% CI, 1.03–1.25; *P*_{trend} = 0.02; **Table 2**). When examining unprocessed red meat and processed meat separately, results were similar, with a slightly stronger association seen for unprocessed red meat than processed meat. Higher poultry intake was statistically significantly associated with lower risk of all-cause mortality (Q4 vs. Q1 RR = 0.90; 95% CI, 0.82–0.98; *P*_{trend} = 0.04). No association was found between any prediagnosis dietary exposure and PCSM (**Table 2**). Processed meat intake had a suggestive but nonsignificant positive association with CVD; poultry was inversely associated with CVD (Supplementary Table S3).

In the postdiagnosis analyses, consistent with results in the prediagnosis analyses, higher intakes of total red and processed meat and poultry were significantly associated with all-cause mortality (Q4 vs. Q1: RR = 1.22; 95% CI, 1.07–1.39; *P*_{trend} = 0.03 for total red and processed meat; RR = 0.84; 95% CI, 0.75–0.95; *P*_{trend} = 0.01 for poultry; **Table 3**). No association was found for PCSM, except that postdiagnostic unprocessed red meat intake was inversely associated with PCSM. Poultry was inversely associated with CVD (Supplementary Table S3).

When we excluded chicken or turkey sandwiches (processed poultry) from the poultry variable, the inverse association of postdiagnosis poultry with all-cause mortality was strengthened (Q4 vs. Q1 RR = 0.80; 95% CI, 0.72–0.90; *P*_{trend} < 0.001). Results were materially the same when adding back men whose prediagnosis survey was

Table 1. Baseline (1992/1993) characteristics according to prediagnosis intakes of meat and eggs among 9,286 men with nonmetastatic prostate cancer in the CPS-II Nutrition Cohort^a.

	Red/processed meat		Poultry		Fish		Eggs	
	Q1	Q4	Q1	Q4	Q1	Q4	Q1	Q4
Participants	<i>n</i> = 2,303	<i>n</i> = 2,315	<i>n</i> = 2,033	<i>n</i> = 2,542	<i>n</i> = 2,081	<i>n</i> = 2,437	<i>n</i> = 1,886	<i>n</i> = 4,607
Median intake (serving/wk)	1.6	9.8	0.6	3.5	0.2	3.0	0	2.0
Age at diagnosis (y) ^a	72.2 (6.1)	71.9 (5.9)	72 (5.9)	72.2 (6.2)	72 (6.0)	72.1 (6.1)	72.2 (6.1)	72.2 (6.0)
Diagnosis year								
1992–1998	744 (32.3)	779 (33.7)	678 (33.3)	820 (32.3)	709 (34.1)	789 (32.4)	637 (33.8)	1,555 (33.8)
1999–2005	1,016 (44.1)	1,020 (44.1)	895 (44)	1,071 (42.1)	924 (44.4)	1,040 (42.7)	801 (42.5)	2,024 (43.9)
2006–2014	543 (23.6)	516 (22.3)	460 (22.6)	651 (25.6)	448 (21.5)	608 (24.9)	448 (23.8)	1,028 (22.3)
Tumor extent								
T1/T2	2,149 (93.3)	2,137 (92.3)	1,880 (92.5)	2,376 (93.5)	1,940 (93.2)	2,269 (93.1)	1,770 (93.8)	4,269 (92.7)
T3/T4	154 (6.7)	178 (7.7)	153 (7.5)	166 (6.5)	141 (6.8)	168 (6.9)	116 (6.2)	338 (7.3)
Gleason score								
2–6	1,164 (50.5)	1,108 (47.9)	1,008 (49.6)	1,250 (49.2)	1,022 (49.1)	1,186 (48.7)	955 (50.6)	2,255 (48.9)
7	544 (23.6)	592 (25.6)	523 (25.7)	632 (24.9)	529 (25.4)	603 (24.7)	445 (23.6)	1,147 (24.9)
8–10	296 (12.9)	308 (13.3)	239 (11.8)	341 (13.4)	256 (12.3)	331 (13.6)	239 (12.7)	623 (13.5)
5–7 ^b	205 (8.9)	203 (8.8)	179 (8.8)	212 (8.3)	183 (8.8)	209 (8.6)	168 (8.9)	380 (8.2)
7–10 ^b	44 (1.9)	40 (1.7)	33 (1.6)	50 (2)	45 (2.2)	45 (1.8)	35 (1.9)	85 (1.8)
Unknown	50 (2.2)	64 (2.8)	51 (2.5)	57 (2.2)	46 (2.2)	63 (2.6)	44 (2.3)	117 (2.5)
Nodal involvement								
No	2,273 (98.7)	2,285 (98.7)	2,006 (98.7)	2,519 (99.1)	2,058 (98.9)	2,415 (99.1)	1,857 (98.5)	4,559 (99)
Yes	30 (1.3)	30 (1.3)	27 (1.3)	23 (0.9)	23 (1.1)	22 (0.9)	29 (1.5)	48 (1)
Race								
White	2,249 (97.7)	2,265 (97.8)	1,985 (97.6)	2,463 (96.9)	2,034 (97.7)	2,374 (97.4)	1,856 (98.4)	4,478 (97.2)
Black	23 (1)	38 (1.6)	26 (1.3)	52 (2)	23 (1.1)	45 (1.8)	12 (0.6)	88 (1.9)
Other/unknown	31 (1.3)	12 (0.5)	22 (1.1)	27 (1.1)	24 (1.2)	18 (0.7)	18 (1)	41 (0.9)
Education								
<High school	74 (3.2)	216 (9.3)	159 (7.8)	125 (4.9)	176 (8.5)	106 (4.3)	70 (3.7)	335 (7.3)
High school graduate	259 (11.2)	548 (23.7)	409 (20.1)	334 (13.1)	484 (23.3)	260 (10.7)	285 (15.1)	812 (17.6)
Some college	457 (19.8)	627 (27.1)	512 (25.2)	597 (23.5)	534 (25.7)	535 (22)	401 (21.3)	1,189 (25.8)
≥College graduate	1,513 (65.7)	924 (39.9)	953 (46.9)	1,486 (58.5)	887 (42.6)	1,536 (63)	1,130 (59.9)	2,271 (49.3)
Family history of prostate cancer								
No	2,195 (95.3)	2,183 (94.3)	1,940 (95.4)	2,405 (94.6)	1,984 (95.3)	2,324 (95.4)	1,788 (94.8)	4,412 (95.8)
Yes	108 (4.7)	132 (5.7)	93 (4.6)	137 (5.4)	97 (4.7)	113 (4.6)	98 (5.2)	195 (4.2)
First course of treatment								
Prostatectomy	625 (27.1)	680 (29.4)	573 (28.2)	742 (29.2)	590 (28.4)	656 (26.9)	532 (28.2)	1,381 (30)
Radiation	49 (2.1)	57 (2.5)	53 (2.6)	43 (1.7)	58 (2.8)	53 (2.2)	47 (2.5)	98 (2.1)
Cryosurgery	852 (37)	814 (35.2)	745 (36.6)	896 (35.2)	723 (34.7)	898 (36.8)	676 (35.8)	1,635 (35.5)
Hormone only/watchful waiting	318 (13.8)	300 (13)	231 (11.4)	363 (14.3)	267 (12.8)	337 (13.8)	258 (13.7)	588 (12.8)
Unknown	459 (19.9)	464 (20)	431 (21.2)	498 (19.6)	443 (21.3)	493 (20.2)	373 (19.8)	905 (19.6)
Any history of prediagnosis PSA testing not leading to prostate cancer diagnosis ^c								
No	283 (12.3)	528 (22.8)	400 (19.7)	389 (15.3)	436 (21)	380 (15.6)	280 (14.8)	919 (19.9)
Yes	1,774 (77)	1,501 (64.8)	1,382 (68)	1,884 (74.1)	1,377 (66.2)	1,767 (72.5)	1,410 (74.8)	3,121 (67.7)
Unknown	246 (10.7)	286 (12.4)	251 (12.3)	269 (10.6)	268 (12.9)	290 (11.9)	196 (10.4)	567 (12.3)
BMI, kg/m ²								
<18.5	8 (0.3)	7 (0.3)	7 (0.3)	4 (0.2)	7 (0.3)	4 (0.2)	5 (0.3)	15 (0.3)
18.5–<25	1,089 (47.3)	706 (30.5)	661 (32.5)	1,012 (39.8)	717 (34.5)	988 (40.5)	894 (47.4)	1,486 (32.3)
25–<30	997 (43.3)	1,213 (52.4)	1,069 (52.6)	1,215 (47.8)	1,048 (50.4)	1,160 (47.6)	842 (44.6)	2,378 (51.6)
≥30	172 (7.5)	363 (15.7)	277 (13.6)	287 (11.3)	280 (13.5)	259 (10.6)	124 (6.6)	688 (14.9)
Unknown	37 (1.6)	26 (1.1)	19 (0.9)	24 (0.9)	29 (1.4)	26 (1.1)	21 (1.1)	40 (0.9)
Smoking status								
Never	707 (30.7)	593 (25.6)	532 (26.2)	810 (31.9)	589 (28.3)	712 (29.2)	581 (30.8)	1,262 (27.4)
Current	71 (3.1)	253 (10.9)	178 (8.8)	124 (4.9)	179 (8.6)	127 (5.2)	71 (3.8)	385 (8.4)
Former	1,327 (57.6)	1,296 (56)	1,194 (58.7)	1,410 (55.5)	1,180 (56.7)	1,393 (57.2)	1,085 (57.5)	2,643 (57.4)
Unknown	198 (8.6)	173 (7.5)	129 (6.3)	198 (7.8)	133 (6.4)	205 (8.4)	149 (7.9)	317 (6.9)
Physical activity, MET-hr/wk								
<3.5	156 (6.8)	358 (15.5)	270 (13.3)	215 (8.5)	325 (15.6)	164 (6.7)	143 (7.6)	569 (12.4)
3.5–<8.75	772 (33.5)	864 (37.3)	776 (38.2)	915 (36)	794 (38.2)	803 (33)	637 (33.8)	1,719 (37.3)
8.75–<17.5	520 (22.6)	443 (19.1)	414 (20.4)	547 (21.5)	397 (19.1)	549 (22.5)	442 (23.4)	945 (20.5)
≥17.5	842 (36.6)	627 (27.1)	553 (27.2)	853 (33.6)	541 (26)	907 (37.2)	649 (34.4)	1,336 (29)
Unknown	13 (0.6)	23 (1)	20 (1)	12 (0.5)	24 (1.2)	14 (0.6)	15 (0.8)	38 (0.8)

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Table 1. Baseline (1992/1993) characteristics according to prediagnosis intakes of meat and eggs among 9,286 men with nonmetastatic prostate cancer in the CPS-II Nutrition Cohort^a. (Cont'd)

	Red/processed meat		Poultry		Fish		Eggs	
	Q1	Q4	Q1	Q4	Q1	Q4	Q1	Q4
History of diabetes								
No	2,183 (94.8)	2,126 (91.8)	1,894 (93.2)	2,367 (93.1)	1,954 (93.9)	2,271 (93.2)	1,773 (94)	4,283 (93)
Yes	120 (5.2)	189 (8.2)	139 (6.8)	175 (6.9)	127 (6.1)	166 (6.8)	113 (6)	324 (7)
History of cardiovascular disease								
No	1,978 (85.9)	2,100 (90.7)	1,827 (89.9)	2,212 (87)	1,840 (88.4)	2,162 (88.7)	1,577 (83.6)	4,204 (91.3)
Yes	325 (14.1)	215 (9.3)	206 (10.1)	330 (13)	241 (11.6)	275 (11.3)	309 (16.4)	403 (8.7)
History of other cancer								
No	2,155 (93.6)	2,169 (93.7)	1,913 (94.1)	2,388 (93.9)	1,960 (94.2)	2,302 (94.5)	1,770 (93.8)	4,331 (94)
Yes	148 (6.4)	146 (6.3)	120 (5.9)	154 (6.1)	121 (5.8)	135 (5.5)	116 (6.2)	276 (6)
Alcohol intake, servings/d								
0	752 (32.7)	777 (33.6)	740 (36.4)	828 (32.6)	875 (42)	590 (24.2)	668 (35.4)	1,390 (30.2)
<2	1,288 (55.9)	1,197 (51.7)	1,028 (50.6)	1,405 (55.3)	961 (46.2)	1,458 (59.8)	988 (52.4)	2,527 (54.9)
≥2	263 (11.4)	341 (14.7)	265 (13)	309 (12.2)	245 (11.8)	389 (16)	230 (12.2)	690 (15)
Fruit and vegetable intake, servings/d	3.5 (1.8)	3.1 (1.5)	2.7 (1.5)	3.6 (1.6)	2.7 (1.5)	3.7 (1.6)	3.4 (1.7)	3.1 (1.5)
Energy intake, kcal/d	1,454.6 (484.5)	2,248.2 (617.0)	1,656 (613.9)	1,913.4 (621.3)	1,712.7 (625.4)	1,911.3 (620.1)	1,596.4 (537.6)	1,931.8 (630.7)

Abbreviation: MET, metabolic equivalent task.

^aNumbers in parentheses are SDs for continuous variables, and percentages for categorical variables.

^bGleason score available as only a categorical score of "intermediate Gleason score" corresponding to scores 5–7 or "high Gleason score" corresponding to scores 7–10.

^cBased on self-reports of PSA screening test reported on 1997 survey when PSA testing was first asked. For cases diagnosed between 1992 and 1997, it was the testing that occurred at least 180 days before diagnosis.

completed within a year of diagnosis date (for all-cause mortality, total red and processed meat Q4 vs. Q1 RR = 1.13; 95% CI, 1.03–1.25; $P_{\text{trend}} = 0.02$; poultry Q4 vs. Q1 RR = 0.90; 95% CI, 0.83–0.98; $P_{\text{trend}} = 0.04$). When excluding men with a CVD history from postdiagnosis analyses, postdiagnosis red/processed meat and total poultry intake were no longer associated with all-cause mortality, while unprocessed poultry was associated with lower risk. Postdiagnosis red meat was still inversely associated with PCSM (Supplementary Table S4). Results did not differ when stratified by age, follow-up time, tumor risk category, BMI, or smoking status (data not shown).

We further examined associations of combinations of dietary intake from before to after diagnosis in relation to all-cause mortality (Table 4). Compared with men who consistently reported intakes below the median, those who consistently reported median or higher intake of total red and processed meat had a higher risk of all-cause mortality (RR = 1.10; 95% CI, 0.99–1.22), primarily driven by processed meat intake. Consistently high intake of poultry was associated with lower risk of all-cause mortality (RR = 0.89; 95% CI, 0.80–0.99).

Substituting 1 serving/wk of poultry for red and/or processed meat was associated with 3% lower risk of all-cause mortality in prediagnosis analysis (all $P < 0.05$) and 3% to 4% lower risk in postdiagnosis analysis (all $P < 0.05$). Substituting 1 serving/wk of poultry for eggs was associated with 4% lower risk of all-cause mortality in both pre- ($P = 0.012$) and postdiagnosis analysis ($P = 0.007$; Fig. 2).

Discussion

In this large cohort study of men with nonmetastatic prostate cancer, higher intake of total red meat and processed meat either before or after cancer diagnosis was associated with higher risk of all-cause mortality. Higher poultry intake either before or after diagnosis was associated with lower risk of all-cause mortality. The effect sizes of relative risk increase or reduction were between 10% and 20%. Furthermore, consistently high intake (median and above) of pro-

cessed meat and poultry before and after diagnosis was associated with higher and lower risk of all-cause mortality, respectively. These associations were largely driven by deaths due to causes other than prostate cancer.

Our finding of a positive association between total red and processed meat intake with overall mortality, primarily due to causes other than prostate cancer, among these nonmetastatic prostate cancer survivors was consistent with findings in studies of more general (noncancer) populations (21, 22). In a dose–response meta-analysis, one serving increase in total red and processed meat intake a day was associated with 17% and 19% increased risk of overall and CVD mortality, respectively (21). Our findings add to the limited evidence suggesting that patients with prostate cancer should adhere to the current nutrition guidelines for cancer survivors, which are consistent with the nutrition guidelines for cancer prevention, i.e., to limit total red and processed meat consumption (4, 23).

Red and processed meat are major sources of animal fat and are also sources of carcinogens. Both types are sources of heterocyclic amines and polycyclic aromatic hydrocarbons produced during high-temperature cooking (24). Processed meat is a rich source of N-nitroso compounds, which can also be formed endogenously. Red meat contains heme, which can catalyze the formation of N-nitroso compounds in the gastrointestinal tract. (25). Red meat and processed meat was classified as "probably carcinogenic to humans" (Group 2A) and "carcinogenic to humans" (Group 1), respectively by International Agency for Research on Cancer in 2015 based on epidemiologic evidence and mechanistic support (26). Although there is some evidence of a moderate positive association between red and processed meat intake and advanced prostate cancer risk in a recent large pooled analysis of 15 cohort studies, the association with incident fatal prostate cancer risk was null (3).

Our findings were generally consistent with previous studies among prostate cancer patients that found null associations between red and processed meat and prostate cancer progression. An analysis of 1,202

Table 2. RRs and 95% CIs for prediagnosis meat and egg intakes and PCSM and all-cause mortality among 9,286 men with nonmetastatic prostate cancer in the CPS-II Nutrition Cohort.

	Quartile of intake ^a				<i>P</i> _{trend}
	Q1	Q2	Q3	Q4	
Total red and processed meat, median (servings/wk)	1.6	3.8	6.0	9.8	
PCSM					
Deaths/person years	147/27,503	167/27,158	190/27,084	162/26,661	
Multivariable model ^b	1.00 (-)	1.07 (0.85-1.34)	1.18 (0.93-1.49)	0.90 (0.69-1.18)	0.41
All-cause mortality					
Deaths/person years	1,087/27,503	1,124/27,158	1,228/27,084	1,243/26,661	
Multivariable model ^b	1.00 (-)	1.07 (0.98-1.16)	1.11 (1.01-1.21)	1.13 (1.03-1.25)	0.02
Unprocessed red meat, median (servings/wk)	1.0	2.0	3.4	5.3	
PCSM					
Deaths/person years	156/27,195	159/27,595	191/28,451	160/25,166	
Multivariable model ^b	1.00 (-)	1.00 (0.79-1.25)	1.11 (0.88-1.39)	1.05 (0.81-1.36)	0.56
All-cause mortality					
Deaths/person years	1,130/27,195	1,150/27,595	1,243/28,451	1,159/25,166	
Multivariable model ^b	1.00 (-)	1.01 (0.92-1.09)	1.07 (0.98-1.16)	1.13 (1.02-1.24)	0.01
Processed meat, median (servings/wk)	0.2	1.2	2.5	5.2	
PCSM					
Deaths/person years	147/26,704	173/29,147	173/25,816	173/26,739	
Multivariable model ^b	1.00 (-)	0.99 (0.79-1.24)	1.07 (0.85-1.36)	0.92 (0.72-1.18)	0.48
All-cause mortality					
Deaths/person years	1,053/26,704	1,201/29,147	1,169/25,816	1,259/26,739	
Multivariable model ^b	1.00 (-)	1.01 (0.93-1.10)	1.09 (1.00-1.19)	1.08 (0.98-1.18)	0.09
Poultry, median (servings/wk)	0.6	1.2	2.0	3.5	
PCSM					
Deaths/person years	146/23,305	228/34,878	117/20,549	175/29,674	
Multivariable model ^b	1.00 (-)	1.01 (0.82-1.26)	0.91 (0.71-1.17)	0.96 (0.76-1.22)	0.63
All-cause mortality					
Deaths/person years	1,095/23,305	1,495/34,878	873/20,549	1,219/29,674	
Multivariable model ^b	1.00 (-)	0.93 (0.86-1.00)	0.95 (0.86-1.04)	0.90 (0.82-0.98)	0.04
Fish, median (servings/wk)	0.2	1.0	1.6	3	
PCSM					
Deaths/person years	161/23,952	163/26,436	180/29,587	162/28,431	
Multivariable model ^b	1.00 (-)	0.92 (0.73-1.15)	0.92 (0.74-1.15)	0.87 (0.69-1.09)	0.26
All-cause mortality					
Deaths/person years	1,127/23,952	1,125/26,436	1,240/29,587	1,190/28,431	
Multivariable model ^b	1.00 (-)	0.93 (0.85-1.01)	0.92 (0.85-1.00)	0.96 (0.88-1.04)	0.54
Eggs, median (servings/wk)	0	0.2	0.6	2.0	
PCSM					
Deaths/person years	127/22,082	61/11,979	129/21,083	349/53,263	
Multivariable model ^b	1.00 (-)	0.94 (0.69-1.28)	1.05 (0.81-1.35)	1.06 (0.85-1.31)	0.52
All-cause mortality					
Deaths/person years	927/22,082	471/11,979	849/21,083	2,435/53,263	
Multivariable model ^b	1.00 (-)	0.97 (0.87-1.09)	0.99 (0.90-1.09)	1.03 (0.95-1.12)	0.18

^aQuartile ranges in 1992 were <2.8, 2.8-4.9, 4.9-7.6, ≥7.6 servings/wk for red/processed meat; <1.6, 1.6-2.7, 2.7-4.2, ≥4.2 servings/wk for unprocessed red meat; <0.8, 0.8-1.8, 1.8-3.6 servings/wk for processed meat; <1.0, 1.0-2.0, 2.0-2.6, ≥2.6 servings/wk for poultry; <0.7, 0.7-1.2, 1.2-2.2, ≥2.2 servings/wk for fish; <0.2, 0.2-0.6, 0.6-1, ≥1 servings/wk for eggs.

^bMultivariable models were adjusted for age at diagnosis; calendar year of diagnosis; tumor extent; Gleason score; nodal involvement; education; family history of prostate cancer; history of PSA testing; BMI; smoking status; physical activity; history of diabetes, cardiovascular disease history, and other cancer; total fruit and vegetable intake; and energy intake. All exposures were adjusted for one another; red and processed meats were mutually adjusted.

men with localized/regional prostate cancer in the Health Professional Follow-up Study (HPFS) found no association between postdiagnosis total red and processed meat intake and risk of prostate cancer progression (94% outcomes were biochemical recurrence; ref. 5). A later analysis in the HPFS among 3,127 men with nonmetastatic prostate cancer reported a suggestive positive association between processed but not red meat intake and risk of lethal prostate cancer (metastasis and PCSM; ref. 8). In the Cancer of the Prostate Strategic Urologic Research Endeavor (CaPSURE) among 1,294 men with localized/regional prostate cancer, intakes of red and processed meat

were not associated with prostate cancer progression (6). A recent analysis of 940 men with nonmetastatic prostate cancer in the Washington University Genetics Study found that intakes of red or processed meat at diagnosis were not associated with prostate cancer progression; however, substituting 30 g/day of poultry or fish for total or red meat was associated with lower risk of progression (7). Outcomes in previous studies primarily included biochemical recurrences that may not eventually metastasize. The present study, with the largest number of PCSM to date, does not support an adverse effect of red and/or processed meat consumption on deaths due to prostate cancer.

Table 3. RRs and 95% CIs for postdiagnosis meat and egg intakes and PCSM and all-cause mortality among 4,882 men with nonmetastatic prostate cancer in the CPS-II Nutrition Cohort.

	Quartile of intake ^a				<i>P</i> _{trend}
	Q1	Q2	Q3	Q4	
Total red and processed meat, median (servings/wk)	2.0	3.8	5.7	9.7	
PCSM					
Deaths/person years	93/12,207	89/12,287	104/11,906	100/12,116	
Multivariable model ^b	1.00 (-)	0.97 (0.71-1.32)	1.11 (0.81-1.52)	0.94 (0.66-1.33)	0.73
All-cause mortality					
Deaths/person years	634/12,207	703/12,287	682/11,906	749/12,116	
Multivariable model ^b	1.00 (-)	1.20 (1.07-1.35)	1.16 (1.03-1.31)	1.22 (1.07-1.39)	0.03
Unprocessed red meat, median (servings/wk)	1.1	2.0	2.9	5.5	
PCSM					
Deaths/person years	98/10,923	101/12,505	94/12,385	93/12,704	
Multivariable model ^b	1.00 (-)	0.86 (0.64-1.15)	0.74 (0.54-1.02)	0.64 (0.46-0.91)	0.01
All-cause mortality					
Deaths/person years	628/10,923	678/12,505	712/12,385	750/12,704	
Multivariable model ^b	1.00 (-)	0.97 (0.86-1.09)	1.05 (0.93-1.19)	1.02 (0.89-1.17)	0.65
Processed meat, median (servings/wk)	0.6	1.3	2.2	5.1	
PCSM					
Deaths/person years	91/11,880	87/12,080	101/12,516	107/12,041	
Multivariable model ^b	1.00 (-)	1.03 (0.75-1.42)	1.16 (0.84-1.60)	1.12 (0.80-1.57)	0.54
All-cause mortality					
Deaths/person years	616/11,880	678/12,080	722/12,516	752/12,041	
Multivariable model ^b	1.00 (-)	1.14 (1.01-1.28)	1.15 (1.02-1.30)	1.17 (1.04-1.33)	0.07
Poultry, median (servings/wk)	0.6	1.1	1.6	4.0	
PCSM					
Deaths/person years	79/9,476	99/11,128	107/14,426	101/13,486	
Multivariable model ^b	1.00 (-)	1.07 (0.78-1.46)	1.04 (0.76-1.42)	1.02 (0.74-1.42)	0.95
All-cause mortality					
Deaths/person years	633/9,476	637/11,128	811/14,426	687/13,486	
Multivariable model ^b	1.00 (-)	0.91 (0.81-1.02)	0.94 (0.84-1.05)	0.84 (0.75-0.95)	0.01
Fish, median (servings/wk)	0.6	1.1	1.6	2.6	
PCSM					
Deaths/person years	85/10,840	104/12,366	97/12,450	100/12,861	
Multivariable model ^b	1.00 (-)	1.10 (0.82-1.48)	1.09 (0.80-1.48)	1.08 (0.78-1.50)	0.74
All-cause mortality					
Deaths/person years	678/10,840	696/12,366	690/12,450	704/12,861	
Multivariable model ^b	1.00 (-)	0.94 (0.84-1.04)	0.98 (0.88-1.10)	0.97 (0.86-1.09)	0.80
Eggs, median (servings/wk)	0.1	0.5	1.0	3.0	
PCSM					
Deaths/person years	51/7,273	71/9,486	71/9,872	193/21,886	
Multivariable model ^b	1.00 (-)	1.06 (0.73-1.54)	1.02 (0.70-1.49)	1.26 (0.89-1.77)	0.08
All-cause mortality					
Deaths/person years	394/7,273	528/9,486	551/9,872	1295/21,886	
Multivariable model ^b	1.00 (-)	0.95 (0.83-1.09)	0.93 (0.81-1.07)	0.98 (0.86-1.11)	0.65

^aQuartile ranges in 1999 were <2.9, 2.9-4.6, 4.6-7.4, ≥7.4 servings/wk for red/processed meat; <1.6, 1.6-2.5, 2.5-3.9, ≥3.9 servings/wk for unprocessed red meat; <0.9, 0.9-1.8, 1.8-3.5, ≥3.5 servings/wk for processed meat; <1, 1-1.5, 1.5-2.5, ≥2.5 servings/wk for poultry; <0.9, 0.9-1.4, 1.4-2.1, ≥2.1 servings/wk for fish; <0.5, 0.5-1, 1-3, ≥3 servings/wk for eggs. Quartile ranges in 2003 were <3.1, 3.1-4.8, 4.8-7.4, ≥7.4 servings/wk for red/processed meat; <1.7, 1.7-2.7, 2.7-3.9, ≥3.9 servings/wk for unprocessed red meat; <1.0, 1.0-1.8, 1.8-3.4, ≥3.4 servings/wk for processed meat; <1, 1-1.5, 1.5-3.2, ≥3.2 servings/wk for poultry; <0.8, 0.8-1.4, 1.4-2.0, ≥2.0 servings/wk for fish; <0.5, 0.5-1, 1-3, ≥3 servings/wk for eggs.

^bMultivariable models were adjusted for age at diagnosis; calendar year of diagnosis; tumor extent; Gleason score; nodal involvement; treatment; education; family history of prostate cancer; prediagnosis history of PSA testing; BMI; smoking status; physical activity; history of diabetes, cardiovascular disease, and other cancer; total fruit and vegetable intake; and energy intake. All exposures were adjusted for one another; red and processed meats were mutually adjusted.

The inverse association between postdiagnosis unprocessed red meat and PCSM is unexpected and it could be due to chance and needs to be confirmed in other studies.

Poultry, especially poultry with skin, is another major source of animal fat and heterocyclic amines. Chicken is the largest source of heterocyclic amines among different meat types (27). In the CaPSURE study (6), men with higher intake of poultry with skin after diagnosis had statistically significantly higher risk of prostate cancer progression,

but there was no association with poultry without skin. In the HPFS (8), a suggestive positive association between total poultry (including processed poultry) and PCSM was observed, which appeared to be driven by sources of poultry other than poultry without skin (e.g., chicken or turkey with skin, chicken or turkey hot dogs, chicken, or turkey sandwiches). However, in the Washington University Genetics Study, there was an inverse association between poultry intake at diagnosis and prostate cancer progression (7). The pooled analysis of

Table 4. RRs and 95% CIs for combinations of pre- and postdiagnosis meat and egg intake and all-cause mortality among 4,882 men with nonmetastatic prostate cancer in the CPS-II Nutrition Cohort.

	Categories ^a			
	Low/low	Low/high	High/low	High/high
Total red and processed meat				
Deaths/person years	945/17,544	519/8,947	392/6,950	912/15,076
Multivariable model ^b	1.00 (-)	1.06 (0.95–1.19)	1.06 (0.94–1.20)	1.10 (0.99–1.22)
Red meat				
Deaths/person years	947/16,963	563/9,648	359/6,465	899/15,441
Multivariable model ^b	1.00 (-)	1.02 (0.91–1.14)	0.97 (0.85–1.10)	1.05 (0.94–1.17)
Processed meat				
Deaths/person years	876/16,850	582/9,896	418/7,109	892/14,661
Multivariable model ^b	1.00 (-)	1.13 (1.01–1.26)	1.15 (1.01–1.29)	1.14 (1.02–1.27)
Poultry				
Deaths/person years	917/14,562	715/12,844	353/6,042	783/15,068
Multivariable model ^b	1.00 (-)	0.95 (0.86–1.05)	0.92 (0.81–1.04)	0.89 (0.80–0.99)
Fish				
Deaths/person years	932/15,361	520/8,882	442/7,844	874/16,429
Multivariable model ^b	1.00 (-)	1.03 (0.92–1.16)	0.96 (0.86–1.08)	0.98 (0.88–1.08)
Eggs				
Deaths/person years	532/9,768	471/8,245	390/6,992	1,375/23,512
Multivariable model ^b	1.00 (-)	0.96 (0.84–1.10)	1.04 (0.91–1.19)	1.02 (0.92–1.14)

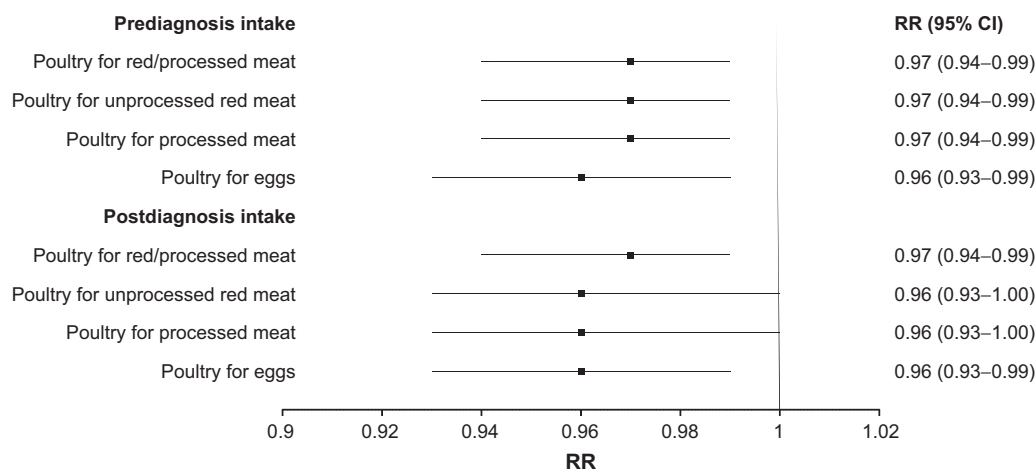
^aLow is defined as below the questionnaire-specific median intake, and high is defined as greater or equal to the questionnaire-specific median intake.

^bMultivariable models were adjusted for age at diagnosis; calendar year of diagnosis; tumor extent; Gleason score; nodal involvement; treatment; education; family history of prostate cancer; prediagnosis history of PSA testing; BMI; smoking status; physical activity; history of diabetes, cardiovascular disease, and other cancer; total fruit and vegetable intake; and energy intake. All exposures were adjusted for one another; red and processed meats were mutually adjusted.

15 cohorts also documented an inverse association between poultry intake and risk of advanced and fatal prostate cancer, respectively (3). The inconsistency may be partly due to different sources of poultry. In contrast, we did not observe any association between poultry intake and PCSM. The null association is unlikely to be explained by the intake level of poultry in this cohort, because the median intake of the top quartile (3.5 servings/wk in prediagnosis analysis and 4 servings/wk in postdiagnosis analysis) was comparable with the median/mean levels of the top quartile among prostate cancer survivors in other studies (6–8). When we further excluded chicken or

turkey sandwiches, considered processed meat, from the poultry variable in postdiagnosis analysis, it was still not associated with PCSM.

However, poultry intake either before or after diagnosis was significantly inversely associated with all-cause mortality, which seems to be driven by sources other than chicken or turkey sandwiches. No previous study investigated poultry intake with all-cause mortality among prostate cancer survivors. Among cancer-free populations, an inverse association between poultry intake and all-cause mortality is suggested in some cohorts (28–30), but not in others (31, 32). Poultry consumption contributes to the overall diet quality in the elderly

**Figure 2.**

RRs and 95% CIs for all-cause mortality associated with substituting 1 serving/wk of poultry for 1 serving/wk of red and/or processed meat or eggs among men with nonmetastatic prostate cancer in the CPS-II Nutrition Cohort. Multivariable models were adjusted for age at diagnosis; calendar year of diagnosis; tumor extent; Gleason score; nodal involvement; treatment (postdiagnosis models only); education; family history of prostate cancer; prediagnosis history of PSA testing; BMI; smoking status; physical activity; history of diabetes, cardiovascular disease, and other cancer; total fruit and vegetable intake; and energy intake.

population and might be the healthy alternative to red and processed meat intake (33). Therefore, future studies that either investigate poultry intake or dietary patterns featuring poultry intake are needed to confirm these findings.

To our knowledge, the current study included the largest number of deaths due to prostate cancer among studies that investigated diet and prostate cancer survival. The long-term follow-up, assessment of both pre- and postdiagnosis diet, and the application of a 2-year lag to minimize reverse causation bias in postdiagnosis analyses also strengthened our findings. The current study also had limitations. The one-time assessment of self-reported pre- and postdiagnosis diet using an FFQ was subject to measurement errors, which could attenuate the estimates of the diet-mortality associations. The first course of treatment was self-reported after cancer diagnosis and was missing among approximately 20% of patients with prostate cancer. Missing treatment information was due to not returning a special postdiagnosis questionnaire, due either to death or nonresponse. Thus, residual confounding due to the missing treatment information may exist. In addition, we had no information on progression or recurrence and only limited information on comorbidities, so potential confounding by these factors is possible. It is worth noting that the majority of the study population were well-educated, older white men; therefore, the study findings may not be generalizable to other populations.

In conclusion, in this large cohort study of men with nonmetastatic prostate cancer, higher intake of total red meat and processed meat either before or after cancer diagnosis was associated with higher risk of overall mortality. Higher poultry intake either before or after diagnosis was associated with lower risk of overall mortality. Our study does not support a positive association between postdiagnosis intakes of red and/or processed meat and PCSM among prostate cancer survivors. However, because the majority of deaths (86%) among prostate cancer survivors in this study were from causes other

than prostate cancer, our findings support current recommendations for prostate cancer survivors to follow nutrition guidelines limiting red and processed meat consumption to improve overall survival.

Disclosure of Potential Conflicts of Interest

No potential conflicts of interest were disclosed.

Authors' Contributions

Conception and design: Y. Wang, R.A. Shah

Development of methodology: Y. Wang, R.A. Shah

Acquisition of data (provided animals, acquired and managed patients, provided facilities, etc.): Y. Wang, E.J. Jacobs

Analysis and interpretation of data (e.g., statistical analysis, biostatistics, computational analysis): Y. Wang, E.J. Jacobs, R.A. Shah

Writing, review, and/or revision of the manuscript: Y. Wang, E.J. Jacobs, R.A. Shah, V.L. Stevens, T. Gansler, M.L. McCullough

Administrative, technical, or material support (i.e., reporting or organizing data, constructing databases): R.A. Shah

Study supervision: M.L. McCullough

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